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Name : Harish kumar JN

Reg: 192524502

Set 11 - Set 15

Set 11

Product Category Analysis

Dataset: Product Categories

Category	Sales (in \$)
Electronics	50000
Clothing	35000
Appliances	40000

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.
2. Build a table to display the sales data for each product category. Label the table elements.
3. Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.
4. Create a pie chart to represent the distribution of sales across product categories. Include labels.

1. Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

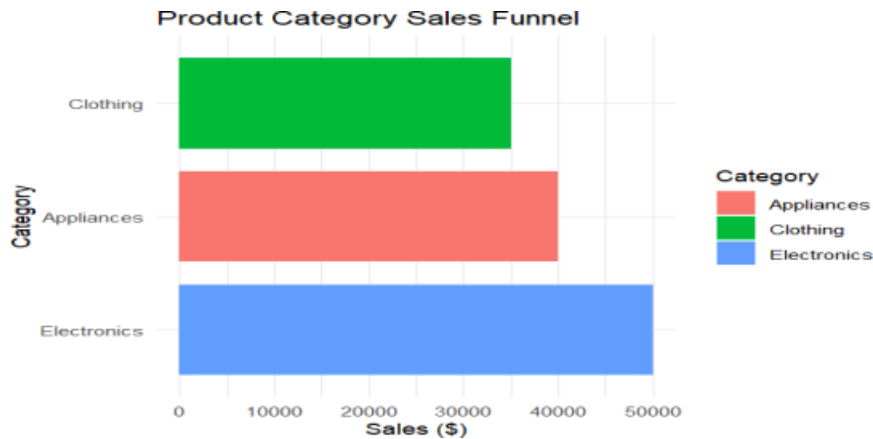
```
# Load library
library(ggplot2)

# Create dataset
category <- data.frame(
  Category = c("Electronics", "Clothing", "Appliances"),
  Sales = c(50000, 35000, 40000)
)

# Funnel Chart
ggplot(category,
  aes(x = reorder(Category, -Sales),
    y = Sales,
    fill = Category)) +
  geom_bar(stat = "identity", width = 0.8) +
  coord_flip() +
  labs(
    title = "Product Category Sales Funnel",
    x = "Category",
    y = "Sales ($)"
```

```
) +
```

```
theme_minimal()
```



2. Build a table to display the sales data for each product category. Label the table elements.

```
# Create dataset
```

```
category <- data.frame(
```

```
Category = c("Electronics", "Clothing", "Appliances"),
```

```
Sales = c(50000, 35000, 40000)
```

```
)
```

```
# Display table
```

```
print(category)
```

```
Category Sales
```

```
1 Electronics 50000
```

```
2 Clothing 35000
```

```
3 Appliances 40000
```

3. Create a pie chart to represent the distribution of sales across product categories. Include labels.

```
# Load library
```

```
library(ggplot2)
```

```
# Create dataset
```

```
category <- data.frame(
```

```
Category = c("Electronics", "Clothing", "Appliances"),
```

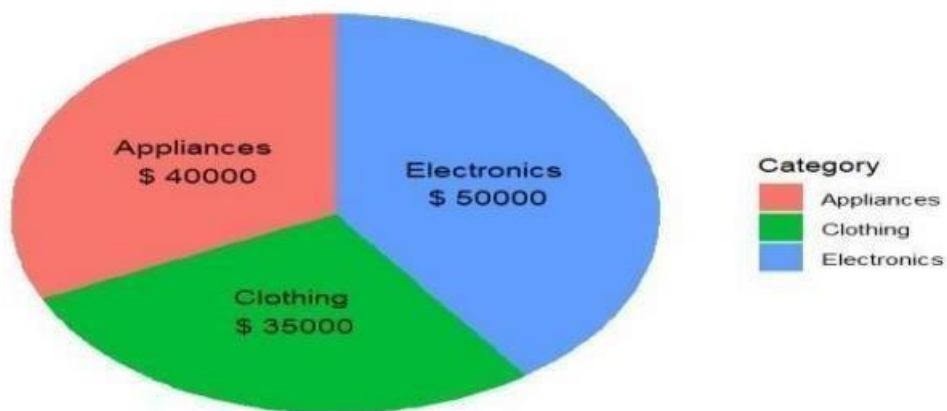
```
Sales = c(50000, 35000, 40000)
```

```
)
```

```
# Pie Chart

ggplot(category, aes(x = "", y = Sales, fill = Category)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste(Category, "\n$", Sales)),
    position = position_stack(vjust = 0.5),
    size = 4
  ) +
  labs(
    title = "Distribution of Sales Across Product Categories"
  ) +
  theme_void()
```

Distribution of Sales Across Product Categories



Set 12

Set 12 :Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%

1. Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.
3. Build a table to show the website traffic data for each day. Label the table elements.
4. Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

1. Create a line chart to visualize the trend in daily page views over time.

```

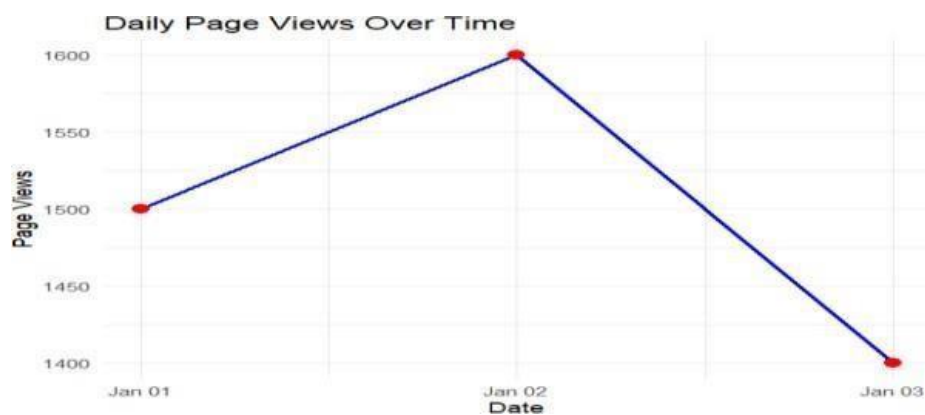
# Load library
library(ggplot2) #

Create dataset

traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Line Chart
ggplot(traffic,
  aes(x = Date,
  y = Page_Views,
  group = 1)) +
  geom_line(color = "blue", linewidth = 1) +
  geom_point(color = "red", size = 3) +
  labs(
    title = "Daily Page Views Over Time",
    x = "Date",
    y = "Page Views"
  ) +
  theme_minimal()

```



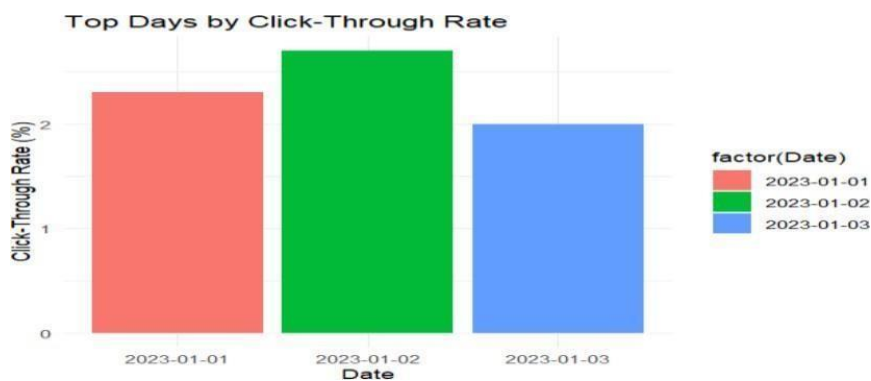
2. Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Load library
library(ggplot2)

# Create dataset
traffic <- data.frame(
  Date = as.Date(c("2023-01-01",
    "2023-01-02",
    "2023-01-03")),
  Click_Through_Rate = c(2.3,2.7,2.0)
)

# Sort data
top_days <- traffic[order(-traffic$Click_Through_Rate), ]

# Bar Chart
ggplot(top_days,
  aes(x = factor(Date),
    y = Click_Through_Rate,
    fill = factor(Date))) +
  geom_bar(stat = "identity") +
  labs(
    title = "Top Days by Click-Through Rate",
    x = "Date",
    y = "Click-Through Rate (%)"
  ) +
  theme_minimal()
```



3. Build a table to show the website traffic data for each day. Label the table elements.

```
# Create dataset
traffic <- data.frame(
  Date = c("2023-01-01",
            "2023-01-02",
            "2023-01-03"),
  Page_Views = c(1500,1600,1400),
  Click_Through_Rate = c("2.3%","2.7%","2.0%")
)

# Display table
print(traffic)
```

```
Date Click_Through_Rate
1 2023-01-01 2.3
2 2023-01-02 2.7
3 2023-01-03 2.0
```

13. Dataset: Geographic Data

1. Create a map chart to visualize the distribution of cities on a geographic map. Label the map elements.

```
library(ggplot2)
library(maps)

geo$Longitude <- c(-120, -100, -80)
geo$Latitude <- c(35, 40, 45)

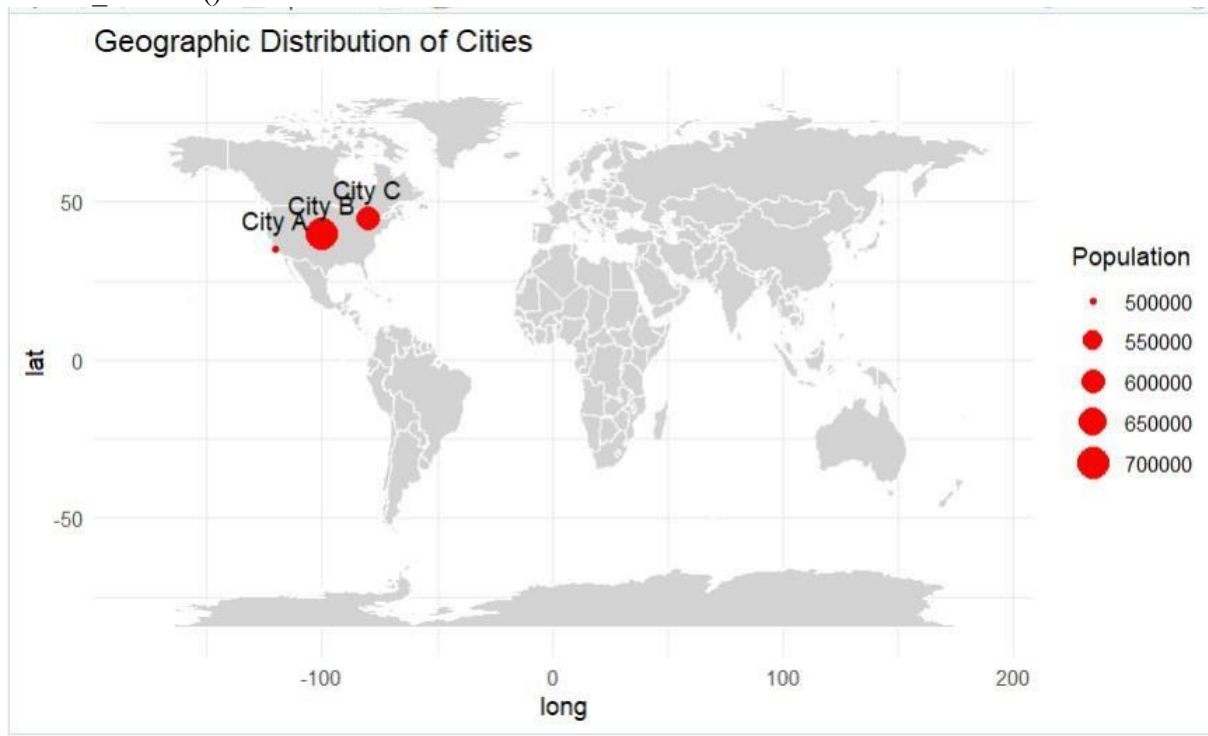
world <- map_data("world")

ggplot() +
  geom_map(
    data = world,
    map = world,
    aes(long, lat, map_id = region),
    fill = "lightgray",
    color = "white"
  ) +
  geom_point(
    data = geo,
    aes(x = Longitude, y = Latitude, size = Population),
    color = "red"
```

```

) +
geom_text(
  data = geo,
  aes(x = Longitude, y = Latitude, label = City),
  vjust = -1
) +
labs(
  title = "Geographic Distribution of Cities"
) +
theme_minimal()

```



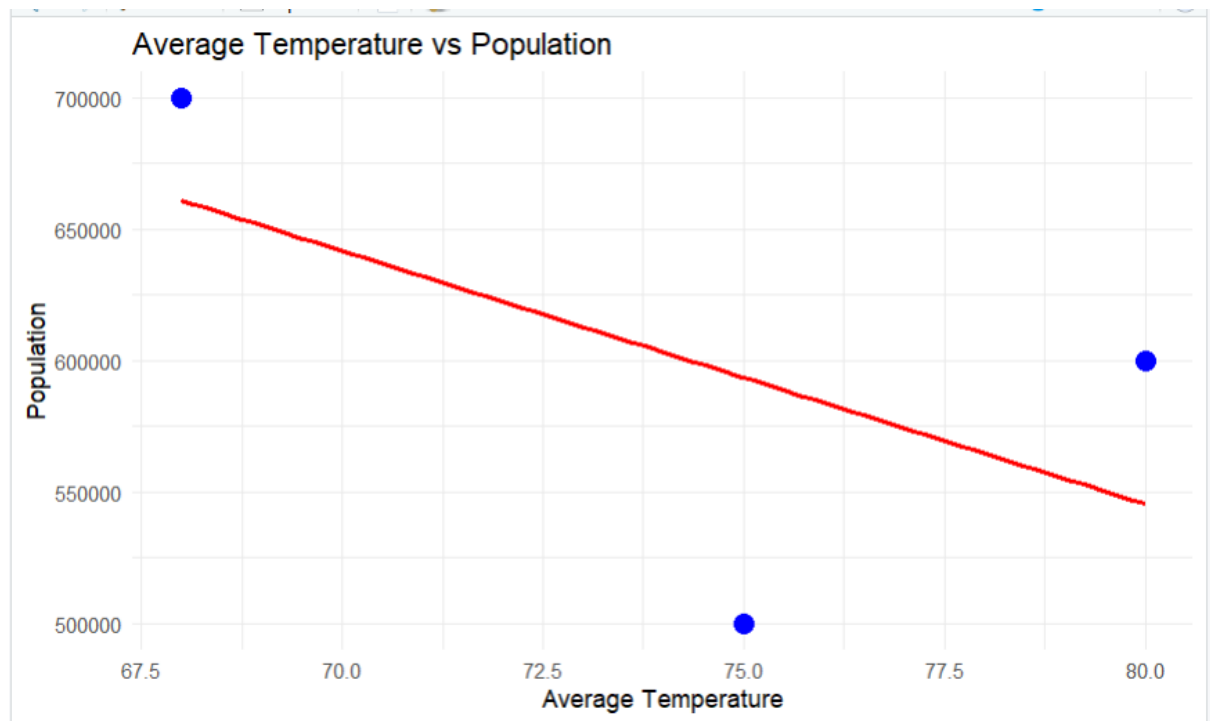
2. Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

```
library(ggplot2)
```

```

ggplot(geographic_data,
  aes(x = Avg_Temperature,
    y = Population)) +
geom_point(
  size = 4,
  color = "blue"
) +
geom_smooth(
  method = "lm",
  se = FALSE,
  color = "red"
) +
labs(
  title = "Average Temperature vs Population",
  x = "Average Temperature",
  y = "Population"
) +
theme_minimal()

```



3. Build a table to display the geographic data for each city. Label the table elements.

Display the geographic data table

View(geographic_data)

	City	Population	Avg_Temperature	Elevation	Longitude	Latitude
1	City A	5e+05	75	1000	-120	35
2	City B	7e+05	68	800	-100	40
3	City C	6e+05	80	1200	-80	45

14. Dataset: Survey Results

Respondent Question 1 Question 2 Question 3

1 A B C

2 B A D

3 C A B

1. Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

library(ggplot2)

library(reshape2)

Convert data to long format

```
long <- melt(survey,
  id.vars = "Respondent",
  variable.name = "Question",
  value.name = "Answer")
```

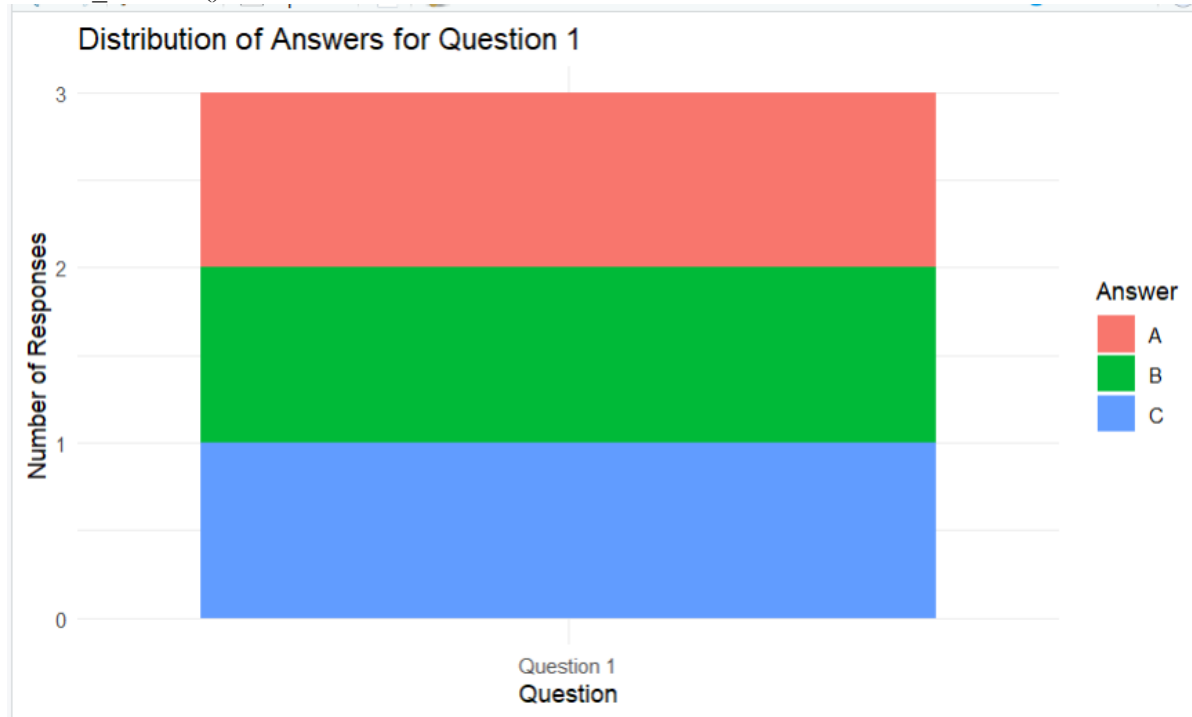
Stacked Bar Chart for Question 1

```
q1 <- subset(long, Question == "Question1")
```

```
ggplot(q1,
  aes(x = "Question 1",
    fill = Answer)) +
  geom_bar() +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
```



```
y = "Number of Responses"
) +
theme_minimal()
```



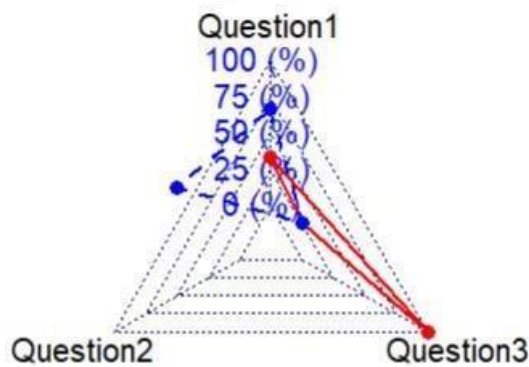
2. Generate a radar chart to visualize the overall pattern of responses across all three questions.

```
score <- data.frame(
  Respondent = survey$Respondent,
  Question1 = c(1,2,3),
  Question2 = c(2,1,1),
  Question3 = c(3,4,2)
)
```

```
radar <- data.frame(
  Question1 = c(4,1,2,3),
  Question2 = c(4,2,1,1),
  Question3 = c(4,3,4,2)
)
```

```
rownames(radar) <- c("Max", "Respondent1", "Respondent2", "Respondent3")
```

```
fmsb::radarchart(
  radar,
  axistype = 1,
  pcol = c("red", "blue", "green"),
  plwd = 2
)
```



3. Build a table to show the survey response data for each respondent. Label the table elements.

View(survey)

or

print(survey)

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

Set 15: Product Sales Analysis

Dataset: Monthly Product Sales

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

1. Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

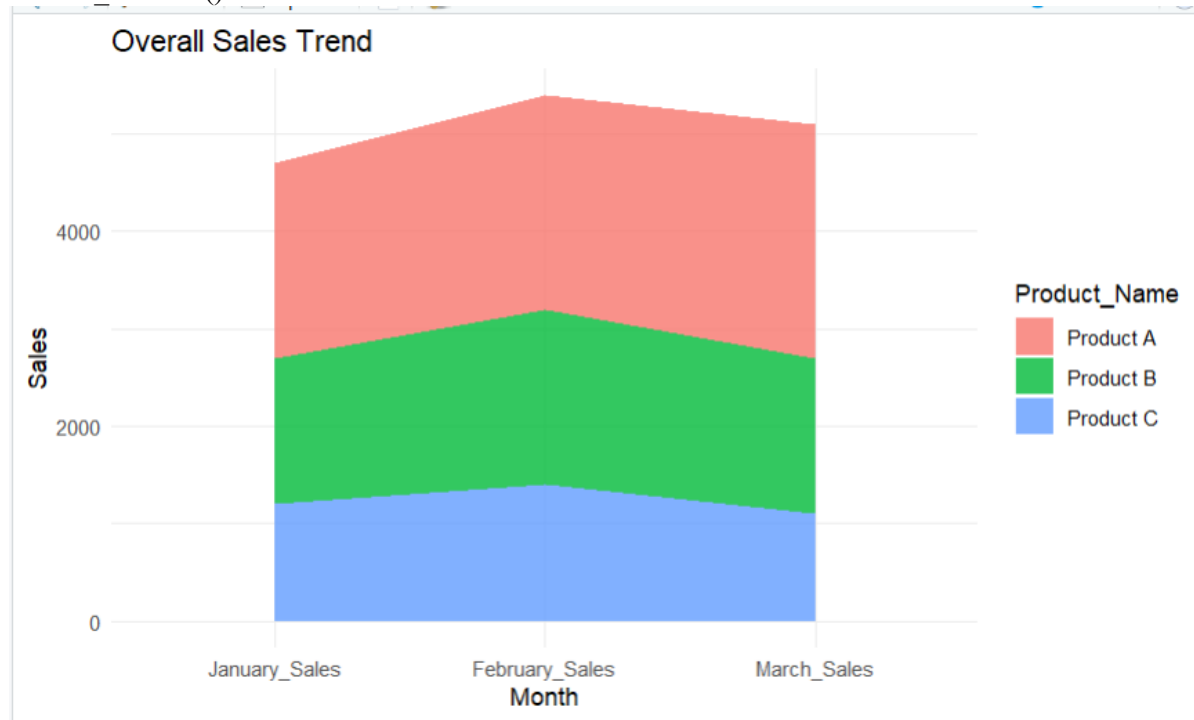
```
long <- melt(
  product_sales,
  id.vars = c("Product_ID", "Product_Name"),
  variable.name = "Month",
  value.name = "Sales"
)
```

```
ggplot(long,
  aes(x = Product_Name,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Quarter 1 Product Sales",
    x = "Product Name",
    y = "Sales"
  ) +
  theme_minimal()
```



2. Generate a stacked area chart to represent the overall sales trend for all products over the three months.

```
ggplot(long,
  aes(x = Month,
    y = Sales,
    fill = Product_Name,
    group = Product_Name)) +
  geom_area(alpha = 0.8) +
  labs(
    title = "Overall Sales Trend",
    x = "Month",
    y = "Sales"
  ) +
  theme_minimal()
```



3. Build a table to show the monthly sales data for each product. Label the table elements.

```
View(product_sales)
```

OR

```
print(product_sales)
```

	Product_ID	Product_Name	January_Sales	February_Sales	March_Sales
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

5 columns

Product_ID

[1, 3]

0% NA

Product_Name

3 unique

0% NA

January_Sales

[1,200, 2,000]

0% NA

February_Sales

[1,400, 2,200]

0% NA

March_Sales

[1,100, 2,400]

0% NA

Showing 1 to 3 of 3 entries, 5 total columns

